

DSA-3 Sample Questions

1) A genome-sequencing startup is building a variant-calling tool that must compare a patient's DNA read against a reference sequence to detect single-nucleotide mutations, insertions and deletions introduced during sequencing. The bioinformatics team wants to quantify how different two sequences are by computing the minimum number of edit operations needed to transform one into the other, and requires an algorithm efficient enough to run on reads that are thousands of bases long.

a) Identify the overlapping subproblems and optimal substructure present in the edit-distance problem, and write down the recurrence relation used by the Wagner-Fischer algorithm.

b) For the sequences $X = \text{"ABBAACBAB"}$ and $Y = \text{"ABBACBAB"}$, apply the Wagner-Fischer algorithm to construct the complete dynamic-programming table, determine the minimum edit distance, and trace back the sequence of edit operations.

c) Distinguish between standard Levenshtein distance, Damerau-Levenshtein distance and weighted edit distance, and justify which variant is best suited to detect sequencing errors caused by adjacent-base swaps.

2) A drone delivery startup must plan a route for a drone that starts at its depot, visits each of a small set of delivery locations exactly once, and returns to the depot, while minimizing the total distance travelled. Since the number of delivery points on any given run is small (at most 15), the operations team decides an exact exponential-time algorithm is feasible instead of a heuristic.

a) Explain the concept of bitmask dynamic programming, describe the state representation $dp[mask][i]$ used for the Travelling Salesman Problem, and derive the resulting time complexity of $O(2^n \cdot n^2)$.

b) For a set of 4 cities with a given distance matrix, apply the Held-Karp bitmask DP algorithm to compute the minimum cost Hamiltonian cycle, clearly showing the DP table over subsets and the optimal tour obtained.

c) Discuss the limitations of bitmask DP as the number of cities grows large, and suggest an alternative approach (heuristic or approximation) that would be more appropriate in that situation.

3) A query planner chooses the cheapest order to multiply a chain of relations. Join Plan models a chain of table-joins as a **matrix-chain product**: the order of association does not change the result but drastically changes the scalar-multiplication (row-scan) cost. It also builds an optimal binary search tree over frequently queried keys the same interval DP shape.

Given matrices: **A1: 50×20** **A2: 20×25** **A3: 25×10** **A4: 10×30**

i.e. $p = [50, 20, 25, 10, 30]$

(a) Fill the interval DP cost table $m[i][j]$ for the chain $A1 \cdot A2 \cdot A3 \cdot A4$ with $p = [50, 20, 25, 10, 30]$. State the minimum number of scalar multiplications and the optimal parenthesisation.

(b) Analyse the structural kinship between matrix-chain multiplication and the optimal binary search tree (OBST) both are interval DPs over a split point k . Explain why a greedy choice (e.g., always multiply the cheapest adjacent pair, or root the most frequent key) fails for both, why the shared recurrence costs $O(n^3)$, and under what condition the Knuth $O(n^2 \log n) / O(n^2)$ optimization applies.

(c) Write the interval DP recurrence $m[i][j] = \min \text{ over } k \text{ of } (m[i][k] + m[k+1][j] + p[i-1] \cdot p[k] \cdot p[j])$, state its base case, and state the order in which cells must be evaluated (by increasing chain length).